

# MATH Unit 4 Previous Year Questions

Mathematics (MATH) · Unit 4 · Discrete Mathematics & Logic · 2024

---

Questions collected from previous year university examinations (2020–2024).

## Part A — Short answer (2 marks)

1. Illustrate Propositional Logic with a worked example. [2022]
2. Define Predicate Logic. Derive or justify the key properties involved. [2022]
3. Discuss the role of Rules of Inference in Discrete Mathematics & Logic. [2024]
4. Write short notes on Set Theory and Relations. [2024]

## Part B — Long answer (10 marks)

1. Discuss the role of Functions in Discrete Mathematics & Logic. [2020]
2. Write short notes on Mathematical Induction. [2022]
3. What is meant by Propositional Logic? Discuss its advantages and limitations. [2021]
4. Illustrate Predicate Logic with a worked example. [2024]
5. Illustrate Rules of Inference with a worked example. [2020]
6. What is meant by Set Theory and Relations? Discuss its advantages and limitations. [2020]
7. Write short notes on Functions. [2021]
8. Discuss the role of Mathematical Induction in Discrete Mathematics & Logic. [2024]
9. Explain Propositional Logic with a neat diagram and a suitable example. [2020]
10. What is meant by Predicate Logic? Discuss its advantages and limitations. [2021]